

1	(a) scalars: energy, power and time	A1	
	vectors: momentum and weight	A1	[2]
	(b) (i) triangle with right angles between 120 m and 80 m, <u>arrows</u> in correct direction and result displacement from start to finish <u>arrow</u> in correct direction and labelled R	B1	[1]
	(ii) 1. average speed ($= 200/27$) $= 7.4 \text{ m s}^{-1}$	A1	[1]
	2. resultant displacement ($= [120^2 + 80^2]^{1/2}$) $= 144 \text{ (m)}$	C1	
	average velocity ($= 144/27$) $= 5.3(3) \text{ m s}^{-1}$	A1	
	direction ($= \tan^{-1} 80/120$) $= 34^\circ$ (33.7)	A1	[3]